Learning When to Trust Retrieval: A Disagreement-Aware Gate for Few-Shot Inverse Dynamics

Constantin Ertel Phillip Kerger
University of California, Berkeley

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Abstract

A visual planner proposes a future camera image, and an inverse-dynamics model (IDM) turns the current image, the robot state and that target image into actions. When a new task comes with only a few demonstrations, retrieving their actions is an attractive way to adapt the IDM. A learned head can combine the retrieved actions with its own prediction, but the gate that decides between the two only sees the robot's state. On a new task where the source-trained prediction is badly wrong, such a gate keeps trusting it, and the learned retrieval head becomes worse than plain nearest-neighbour regression. We give the gate two extra inputs: how confident the retrieval is and how strongly the head's own prediction disagrees with the retrieved actions. During source training we also replace a quarter of the head's predictions with predictions for other inputs, so the gate learns to fall back to retrieval when the two disagree. We designed the method on the ten LIBERO-Goal tasks, froze all methods, settings and three hypotheses, and then evaluated once on 30 sealed tasks from LIBERO-10, LIBERO-Spatial and LIBERO-Object, with three frozen image encoders (MAE, DINOv2, SigLIP), five demonstrations per task and three seeds. Without any target training, the disagreement-aware head had lower action error than k-nearest-neighbour regression on all 30 tasks for every encoder (MAE 0.378 vs 0.528 chunk MSE) and lower error than the same head without the new inputs (0.378 vs 0.425); both pre-registered comparisons held for all three encoders. After 25 head updates it beat head-only fine-tuning for DINOv2 and SigLIP but not for MAE. Fine-tuning the encoder reached lower error (0.287 with full fine-tuning) at 7 to 17 times the compute per update. In closed-loop simulation with recorded target frames, the head adapted for 25 updates succeeded in 38% of episodes against 16% for head-only fine-tuning and 31% for LoRA, but plain k-nearest-neighbour regression succeeded in 44%, and lower offline error did not predict higher success.
Keywords: inverse dynamics; retrieval; gating; few-shot adaptation; robot manipulation; LIBERO.

1 Introduction

Visual planning splits robot control into two parts: a generative model proposes what the scene should look like next, and an inverse-dynamics model (IDM) infers the actions that produce that change. UniPi introduced this split for text-conditioned video planning [1], and recent systems keep it, from visual subgoals in $\pi 0.7$ [2] to the action modules of GE-Act 2.0 and VPP2 [3, 4]. When the planner is fixed, adapting to a new task means adapting the IDM, and a new task usually comes with only a few demonstrations.

The simplest way to use a few demonstrations is to look up the closest recorded transition and copy its actions, as in VINN [5]. A learned head can do better by attending over several retrieved candidates and mixing the result with its own prediction through a gate [6, 7]. We

found a failure of this design in an earlier study. Our attention-and-gate head (RAID v7) was better than nearest-neighbour regression on average, but on one held-out LIBERO-Goal task its error was 1.7 times that of nearest-neighbour regression (0.682 vs 0.406). Its errors followed the source-trained prediction closely (Pearson 0.97 across query demonstrations), not the retrieved actions. The gate only saw the robot's state, so it had no way to notice that the source-trained prediction was wrong for this task.

This paper fixes that failure and tests the fix at scale. RAID v8 gives the gate the information it was missing: the confidence of the retrieval and the per-dimension disagreement between the direct prediction and the retrieved prior. A disagreement input alone is not enough, because during source training the direct prediction is rarely far off, so the gate never learns what to do when it is. We therefore corrupt the direct prediction during training: for a quarter of the rows, the mix receives another input's prediction. The gate is then trained on exactly the situation it must handle on a new task: the direct prediction disagrees with confident retrieval.

We evaluate the fix with a protocol that separates design from evaluation. All design choices, including the gate variant, learning rates and the closed-loop gripper rule, were made on the ten LIBERO-Goal tasks. We then froze the method set, settings, analysis code and three hypotheses in a digest-bound amendment, and ran the frozen benchmark once on 30 sealed tasks from three other LIBERO suites [8], with three image encoders: MAE [9], DINOv2 [10] and SigLIP [11].

Our contributions:

1. **A disagreement-aware retrieval gate, trained with direct-path corruption.** Without target training it has lower action error than nearest-neighbour regression on all 30 sealed tasks for each of three encoders, and lower error than the same head without the new inputs (Section 6.1). On development tasks, the gate inputs without corruption training make the head worse, not better (Section 6.2).
2. **A compute-aware comparison with fine-tuning.** Head updates on frozen features cost 4 to 7 ms each; LoRA and full fine-tuning cost 52 and 128 ms. The disagreement-aware head reaches its best error within 25 updates, but encoder fine-tuning reaches lower error with more compute (Section 6.3).
3. **A closed-loop test without a generator,** which replays recorded target frames in the LIBERO simulator. It shows that the adapted disagreement-aware head finishes more tasks than head-only fine-tuning, LoRA and full fine-tuning, that nearest-neighbour regression still finishes the most, and that lower offline error does not predict success (Section 6.4).

Figure 1 summarizes the model and the protocol.

2 Related Work

Video planning with inverse dynamics. UniPi trains a separate inverse model to execute generated videos [1]. π0.7 conditions a generalist policy on visual subgoals [2], and GE-Act 2.0 and VPP2 pair a visual predictor with an action module [3, 4]. Predictive inverse-dynamics models scale this idea [12], and the label efficiency of IDMs depends strongly on the setting [13]. We keep the planner out of the evaluation and study only how the IDM adapts to a new task.

Retrieval for action prediction. VINN predicts actions by locally weighted nearest-neighbour regression on frozen features [5]. REGENT retrieves demonstrations and blends the nearest action with a learned policy through a distance-based weight [6]; nearest-neighbour language models interpolate a parametric and a retrieval distribution in the same way [7]. ReCAP adds trajectory retrieval to a frozen policy to cover new tasks [14]. These methods fix the mixing rule or tie it to distance; our gate is learned and sees the disagreement between the two predictions.

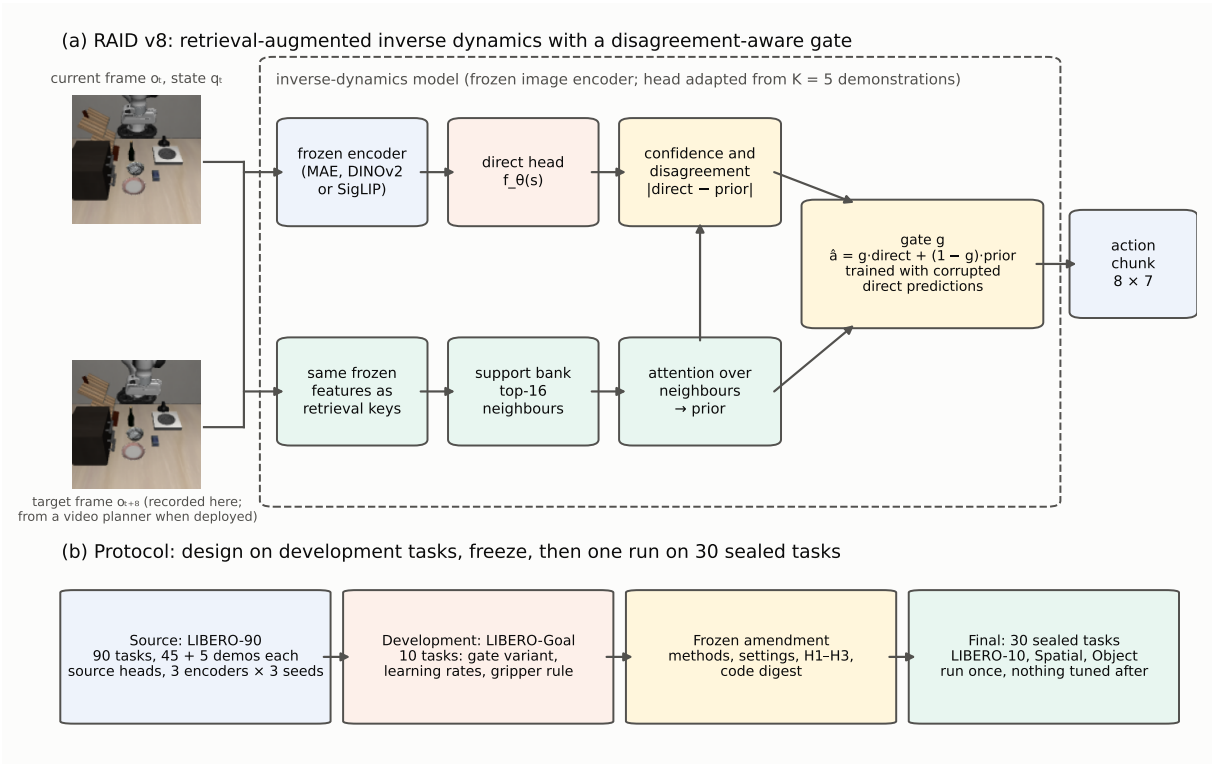


Figure 1: (a) RAID v8. A frozen encoder gives the controller state and the retrieval keys; a direct head predicts an action chunk, attention over the 16 nearest support windows forms a retrieval prior, and a gate mixes the two using the retrieval confidence and the disagreement between them. The gate is trained with corrupted direct predictions. Example frames are from a LIBERO-Goal demonstration. (b) Protocol: source heads on LIBERO-90, design on LIBERO-Goal, a frozen amendment, and one run on 30 sealed tasks.

Retrieval for training data and lightweight adaptation. Behavior Retrieval and STRAP retrieve prior data to train few-shot policies [15, 16]; HAND retrieves and then fine-tunes [17]. LoRA [18] and task-specific adapters [19] adapt large encoders cheaply, and VLA-Pro retrieves and fuses task-specific adapters [20]. We compare against head-only, LoRA and full fine-tuning on the same support sets and report their compute.

Learning when to defer. Learning to defer trains a model to hand a decision to another predictor when it is likely to be wrong [21]. Our gate makes a soft, per-dimension version of this decision between a parametric head and retrieval, and our corruption training supplies the examples of a wrong parametric prediction that ordinary training lacks.

End-to-end few-shot policies. Policies such as Octo and DISC adapt an entire visuomotor policy and report task success; in DISC's five-demonstration LIBERO-Goal evaluation, success reaches 55% for Octo and 73% for DISC [22, 23], under a protocol different from ours. We study a component of a modular pipeline and do not compete with these systems on task success.

3 Problem Setting

Inverse dynamics. At time t the IDM receives the current RGB image o_t, the target image o_{t+8}, the robot state q_t (7 joint and 2 gripper positions) and predicts the next 8 actions $\hat{a}_{t:t+7} \in \mathbb{R}\hat{}\{8\times7\}$ in LIBERO's action space. Actions and states are normalized with statistics of the source training data.

Few-shot adaptation. A new task provides $K = 5$ demonstrations, the support set. A method may use them for retrieval, for parameter updates, or both. It is evaluated on 20 other demonstrations of the same task, the query set.

Offline metric (primary). Every stride-1 window of each query demonstration is evaluated with its recorded target image and recorded action chunk. The error of a window is the mean squared error over its 56 normalized action values; errors are averaged per query demonstration and then over the 20 demonstrations. We also report R^2, the mean over the 7 action dimensions of $1 - \mathrm{SSE/SST}$. Because the target image is recorded, the metric measures action inference given a correct target and does not depend on any generator.

Closed-loop metric (secondary). The simulator starts from the recorded first state of a query demonstration. Before each 8-step chunk, the current image is matched to the closest recorded frame of that demonstration (cosine distance of frozen features, searched forward over 24 frames, never backwards), and the recorded frame 8 steps later becomes the target. The predicted chunk is executed, and success is LIBERO's task predicate within 1.5 times the demonstration length plus 20 steps. This isolates the IDM from planner errors.

Compute. Adaptation GPU-seconds are the summed wall-clock time of parameter updates on one GPU, without evaluation; we also report time per update and trainable parameters. One-time costs shared by all methods (feature caching and source training) are reported separately.

4 Method

4.1 Retrieval-augmented inverse dynamics

A frozen image encoder φ maps each image to a 768-dimensional feature. The controller state is $s = [\varphi(o_t), \varphi(o_{t+8}), q_t, H]$. A support bank stores, for every window of the support demonstrations, the frozen features of its current and target images, its state, its normalized action chunk and its demonstration. Retrieval uses the distance

$$d = 0.5\,(1 - \cos(\varphi(o_t), \varphi(o'_t))) + 0.5\,(1 - \cos(\varphi(o_{t+H}), \varphi(o'_{t+H}))) + 0.1 \cdot \mathrm{mean}((q_t - q'_t)^2),$$

and returns the k = 16 closest windows. Because the encoder stays frozen in all head methods, the retrieval geometry never changes during adaptation.

4.2 The attention head and its failure

The RAID v7 head computes a direct prediction from an MLP trunk, an attention-weighted prior over the 16 retrieved action chunks, and a per-dimension gate g = σ(W_g s) that mixes them; an auxiliary loss pulls attention towards candidates whose actions match the target. The gate is a function of the state s only. If the source-trained trunk is wrong for a new task, nothing in s reveals it, so the gate cannot move towards the prior. This is the failure we observed.

4.3 The disagreement-aware gate

RAID v8 keeps the trunk, the attention and the selection loss of v7 and changes the gate:

$$\text{direct} = f_\theta(s), \qquad \text{prior} = \textstyle\sum_i \alpha_i a^i,$$

where α_i is the attention weight of candidate i and a^i its action chunk. The gate sees a confidence vector

$$c = \left[\, \text{sim}_{(1)},\ \overline{\text{sim}},\ \tilde{H}(\alpha),\ |\text{direct} - \text{prior}|\, \right],$$

with the absolute disagreement taken per action element and detached from the gradient, and computes g = σ(MLP_g([s, c])) $\in (0, 1)^{56}$, with a 256-unit hidden layer, and $\hat{a}$ = g $\odot$ direct + (1 − g) $\odot$ prior. Without any valid candidate the gate is fixed at 1.

4.4 Direct-path corruption

The new inputs only help if the gate has seen large disagreements during training. On source tasks the trunk is usually right, so we create disagreements on purpose: in each training batch, each row's direct prediction is replaced, with probability 0.25 and inside the mix only, by the detached direct prediction of a random other row. For these rows the correct action is closer to the prior, and the loss teaches the gate to take it when the disagreement is large and retrieval is confident. The trunk receives no gradient from corrupted rows. Corruption stays on during target adaptation.

4.5 Source training

One head per type, encoder and seed is trained on the 90 LIBERO-90 tasks (45 training and 5 validation demonstrations each) for 4,000 updates. Each batch draws 8 tasks; for each, an episodic bank of 2, 5 or 10 training demonstrations and 16 query windows from other training demonstrations of that task. The loss averages the retrieval-present error and the error with retrieval removed, plus 0.1 times the selection loss (AdamW, learning rate $3{\cdot}10^{-4}$, weight decay 10^{-4}, gradient clipping 1.0). The checkpoint with the lowest validation error, checked every 250 updates, is kept. With retrieval removed, the same network is a plain MLP, which serves as the starting point of head-only fine-tuning.

4.6 Target adaptation

Head methods update the whole head on frozen features with batch 64 for up to 200 updates. Retrieval-based heads retrieve from the five support demonstrations; during adaptation each support window excludes its own demonstration from the bank, and at evaluation the full bank is used.

5 Experimental Protocol

5.1 Data and splits

We use the LIBERO datasets [8] from their public mirror. Source: all 90 LIBERO-90 tasks. Development: the 10 LIBERO-Goal tasks, which our earlier work had already used and which we therefore used only for design. Final: the 30 tasks of LIBERO-10, LIBERO-Spatial and LIBERO-Object, all of them, none selected; their files were downloaded and read only in the final run. LIBERO-10 shares scenes with LIBERO-90 but has different tasks; LIBERO-Spatial and LIBERO-Object use other layouts and objects. For each target task, a seeded permutation of its 50 demonstrations gives a support pool of 30 and a query set of 20, and support seeds 42, 43 and 44 each draw $K = 5$ support demonstrations.

5.2 Methods

Table 1 lists the compared methods. Head-only fine-tuning uses the same source network as RAID v8 with retrieval off, so the two differ only in retrieval. LoRA (rank 8, $\alpha = 16$, query-key-value and output projections of the last two MAE blocks) and full fine-tuning update the encoder and the head, are run with MAE and seed 42 only, and use retrieval off. All learning rates were chosen on development tasks from three-value grids.

Table 1. Methods. Trainable parameters are those updated on the target task.

Method	Retrieval	Updated on target	Learning rate	Trainable params
VINN-style k-NN (k = 4)	yes	none	—	0
REGENT-style blend	yes	none	—	0
RAID v7 head	yes	head	10^{-4}	1,420,832
RAID v8 head (ours)	yes	head	10^{-4}	1,759,728
Head-only fine-tuning	no	head	$3{\cdot}10^{-4}$	1,759,728
LoRA (MAE)	no	adapter and head	$3{\cdot}10^{-4}$	1,833,456
Full fine-tuning (MAE)	no	encoder and head	10^{-5}	87,558,384

5.3 Pre-registration and statistics

After development we froze the method set, all settings, the analysis code and three hypotheses in an amendment (SHA-256 digest 84375c3b…), and the final run checked this digest before reading any final file. The hypotheses, tested separately for each encoder with MAE as the primary encoder:

- **H1:** without target training, RAID v8 has lower chunk MSE than RAID v7.
- **H2:** without target training, RAID v8 has lower chunk MSE than the best training-free baseline on development tasks, which was k-NN.
- **H3:** after 25 updates, RAID v8 has lower chunk MSE than head-only fine-tuning after 25 updates.

For each task we average the paired difference over the three seeds. The 95% interval is a percentile bootstrap over the 30 tasks (10,000 resamples), the one-sided p-value is the share of bootstrap means at or above zero, and Holm's correction is applied over H1–H3. A hypothesis is supported if its Holm-adjusted p-value is below 0.05 and its interval lies entirely below zero. Everything else is descriptive.

5.4 Closed loop and the gripper rule

The closed-loop test runs 20 query episodes per final task for k-NN, RAID v7 and RAID v8 without training, head-only and RAID v8 after 25 updates, and LoRA and full fine-tuning after 200 updates (seed 42, MAE). On development tasks we saw that learned heads output gripper commands between open and closed, which often fails to grasp; rounding the gripper command to fully open or fully closed raised head-only success on a development pick-and-place task from 30% to 60% and k-NN from 90% to 100%. The frozen protocol therefore rounds the executed gripper command for every method. No model changes.

6 Results

6.1 Without target training, the disagreement-aware gate beats nearest-neighbour regression on every task

Table 2. Final tasks, no target training: chunk MSE (mean over 30 tasks × 3 seeds) and R^2. Lower MSE and higher R^2 are better.

Method	MAE	DINOv2	SigLIP	R^2 (MAE)
Source head, retrieval off	0.683	0.655	0.763	0.05
VINN-style k-NN	0.528	0.421	0.448	0.25
REGENT-style blend	0.554	0.388	0.437	0.22
RAID v7	0.425	0.447	0.446	0.38
RAID v8 (ours)	**0.378**	**0.348**	**0.364**	**0.46**

H1 and H2 were supported for all three encoders (Figure 3). RAID v8 had lower error than k-NN on all 30 tasks for every encoder (MAE −0.151, 95% CI [−0.169, −0.133]; DINOv2 −0.073 [−0.088, −0.059]; SigLIP −0.084 [−0.097, −0.072]), and even its worst task was better than k-NN (Figure 2a). It also had lower error than the REGENT-style blend on all 30 tasks for every encoder. Against RAID v7 it was better on 26, 30 and 29 of 30 tasks (MAE −0.047 [−0.062, −0.034]; DINOv2 −0.099 [−0.124, −0.075]; SigLIP −0.082 [−0.100, −0.064]); on the few tasks where v7 was better, the difference was at most 0.011 (Figure 2b).

6.2 The gate needs corruption training

The development tasks contained the ablation that motivated the design (Figure 4, Appendix B). Adding the confidence and disagreement inputs without corruption training made the head worse than RAID v7 for all three encoders (MAE 0.757 vs 0.637). With corruption training, the same head was best (0.494, against 0.592 for k-NN). Without examples of a wrong direct prediction, the gate learns to trust the trunk whenever it is confident; the corrupted rows teach it the opposite case. On the development task where RAID v7 had failed in our earlier study (wine bottle on top of the cabinet), the MAE error was 1.192 for RAID v7, 0.705 for RAID v8 and 0.643 for k-NN.

6.3 With a few updates: a small advantage, and a compute trade-off

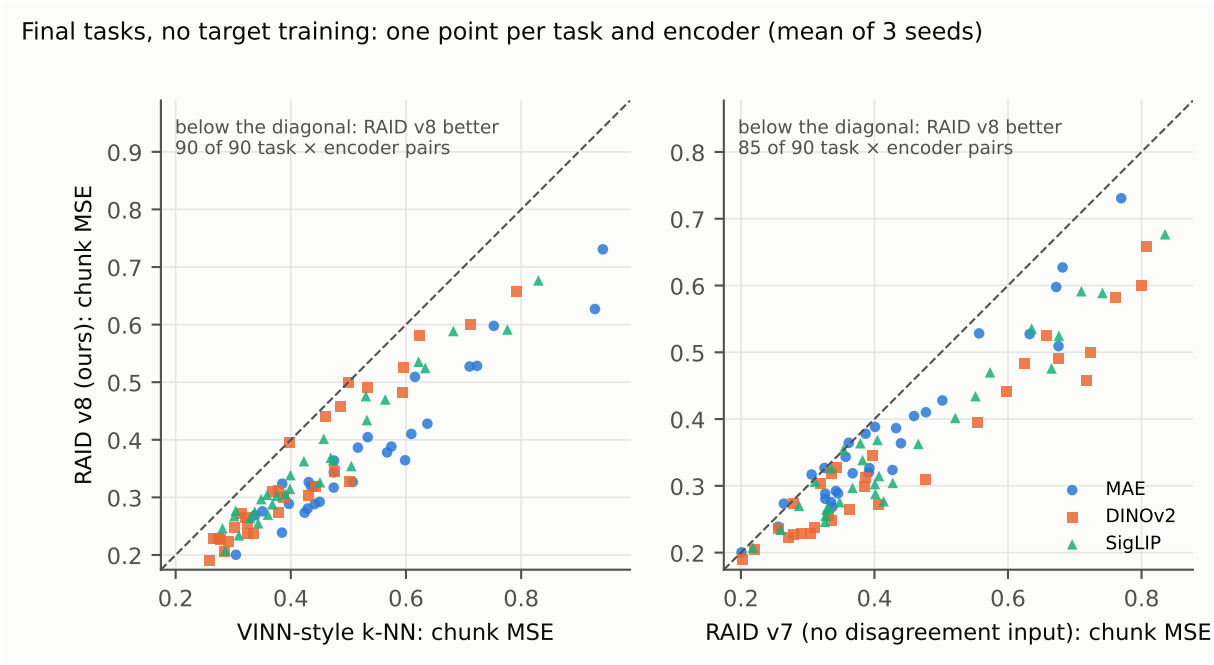


Figure 2: Final tasks, no target training: RAID v8 error for each task and encoder (mean of three seeds) against k-NN (a) and against RAID v7 (b). Points below the dashed line favour RAID v8.

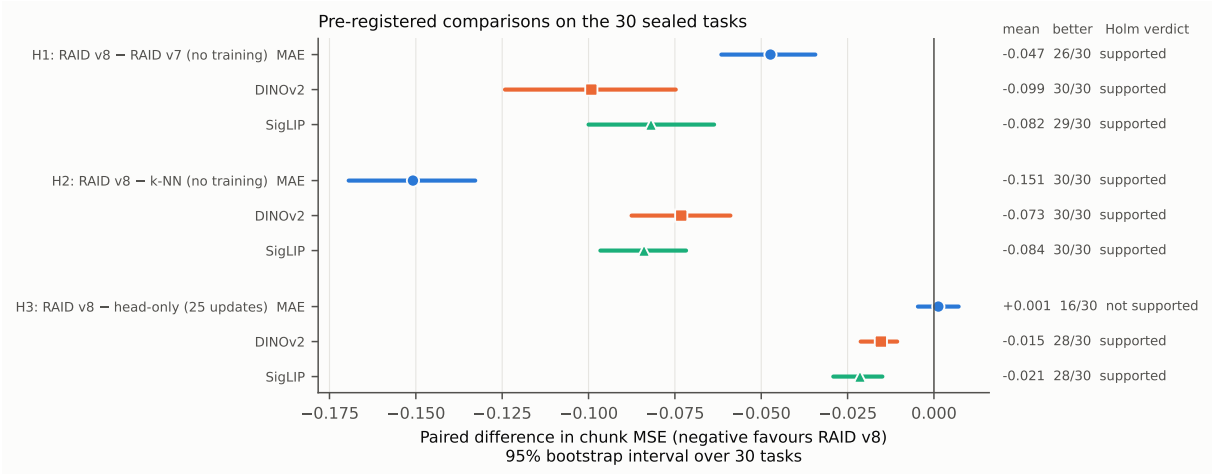


Figure 3: The three pre-registered comparisons for each encoder: paired difference in chunk MSE with a 95% bootstrap interval over the 30 tasks, the number of tasks favouring RAID v8, and the Holm-corrected verdict.

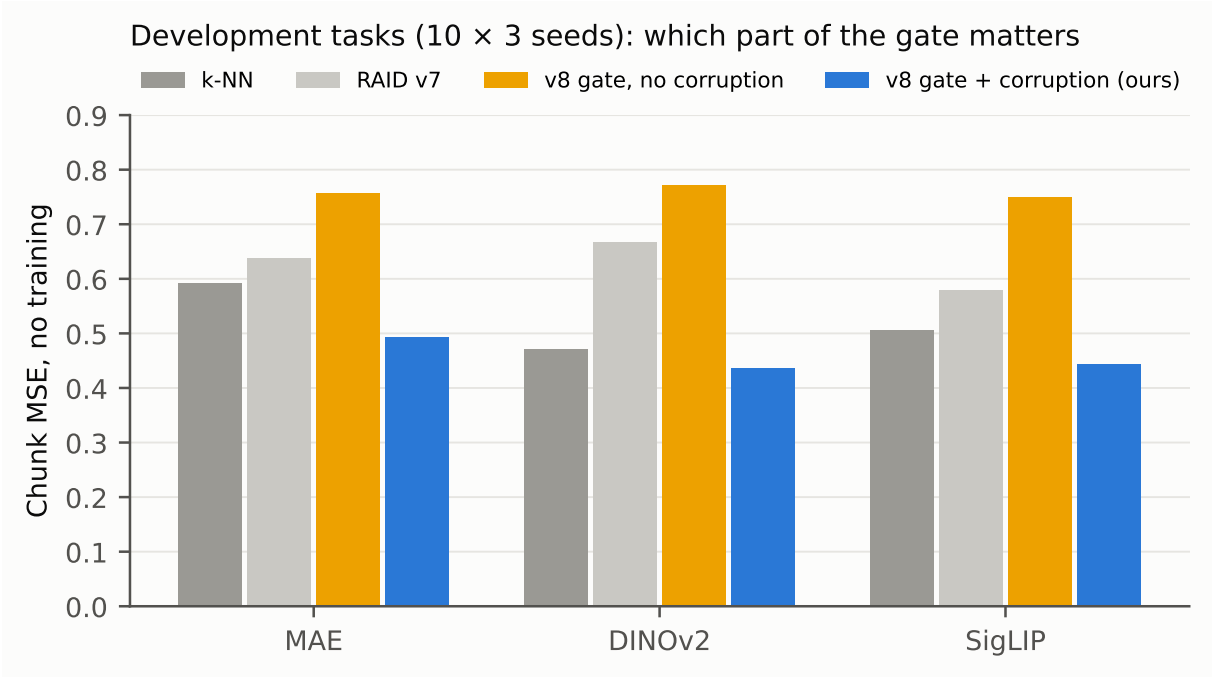


Figure 4: Development tasks, no target training: chunk MSE of k-NN, RAID v7, the v8 gate without corruption training and the v8 gate with corruption training (10 tasks × 3 seeds per encoder). These are design data, not held-out results.

Table 3. Final tasks: chunk MSE after n target updates (mean over 30 tasks × 3 seeds; encoder tuning: MAE, seed 42) and the time per update on an NVIDIA A100 80GB PCIe.

Method	Encoder	0	25	100	200	ms per update
Head-only fine-tuning	MAE	0.683	0.339	0.316	0.318	3.6
RAID v7 head	MAE	0.425	0.323	0.314	0.316	6.9
RAID v8 head (ours)	MAE	0.378	0.341	0.332	0.340	7.3
LoRA	MAE	0.693	0.335	0.300	0.293	51.9
Full fine-tuning	MAE	0.693	0.341	0.288	0.287	127.5
Head-only fine-tuning	DINOv2	0.655	0.323	0.308	0.307	—
RAID v8 head (ours)	DINOv2	0.348	0.308	0.313	0.328	—
Head-only fine-tuning	SigLIP	0.763	0.361	0.346	0.342	—
RAID v8 head (ours)	SigLIP	0.364	0.339	0.352	0.366	—

H3 was supported for DINOv2 (−0.015 [−0.021, −0.011], 28 of 30 tasks) and SigLIP (−0.021 [−0.029, −0.015], 28 of 30) but not for MAE (+0.001 [−0.005, +0.007], 16 of 30). Two further observations are descriptive. First, after updates RAID v7 had slightly lower error than RAID v8 for all three encoders (MAE 0.323 vs 0.341 at 25 updates): the gate's preference for retrieval, which helps without training, limits what head updates can add. Second, RAID v8 reached its lowest error within 25 to 100 updates and then rose slightly, while head-only fine-tuning changed little after 100 updates.

Encoder tuning reached the lowest error: full fine-tuning 0.287 and LoRA 0.293 after 200 updates, against 0.316 for the best head method. It also costs more. One update cost 7.3 ms for the RAID v8 head, 52 ms for LoRA and 128 ms for full fine-tuning, and 200 updates took 1.3, 10.5 and 25.8 GPU-seconds (Figure 5). At 25 updates the three reached similar error (0.341, 0.335 and 0.341) after 0.17, 1.3 and 3.2 GPU-seconds. The one-time costs are shared by all methods: encoding all frames of the 30 final tasks with three encoders took 7.6 minutes, and training one source head took about 2.8 minutes on an A100.

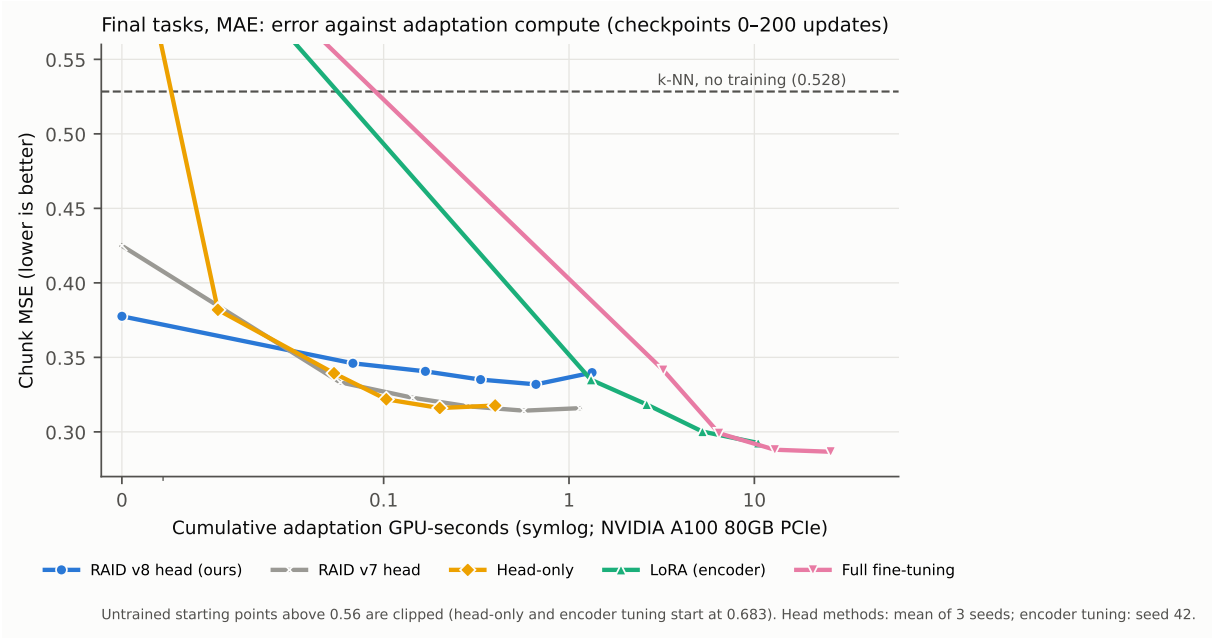


Figure 5: Final tasks, MAE: chunk MSE against cumulative adaptation GPU-seconds. RAID v8 starts lowest; encoder tuning reaches the lowest error with one to two orders of magnitude more GPU-seconds.

6.4 Closed loop: the adapted head finishes more tasks than fine-tuning, but k-NN finishes the most

Table 4. Closed-loop success with recorded target frames (20 episodes per task, seed 42, MAE, rounded gripper), with a 95% bootstrap interval over the 30 tasks.

Method	LIBERO-10	LIBERO-Spatial	LIBERO-Object	All 30 tasks	95% CI
VINN-style k-NN	0.23	0.52	0.58	**0.44**	[0.35, 0.53]
RAID v7, no training	0.06	0.07	0.01	0.05	[0.02, 0.09]
RAID v8, no training	0.08	0.12	0.32	0.17	[0.10, 0.26]
Head-only, 25 updates	0.03	0.40	0.07	0.16	[0.08, 0.26]
RAID v8, 25 updates	0.12	0.58	0.44	0.38	[0.27, 0.50]
LoRA, 200 updates	0.15	0.57	0.21	0.31	[0.22, 0.40]
Full fine-tuning, 200 updates	0.10	0.40	0.30	0.27	[0.20, 0.33]

The disagreement-aware gate also helped in closed loop: without training, RAID v8 succeeded in 17% of episodes against 5% for RAID v7 (paired difference +0.125, 95% CI [+0.04, +0.22], better on 18 tasks and worse on 5). After 25 updates, RAID v8 succeeded in 38% of episodes against 16% for head-only fine-tuning (+0.22 [+0.10, +0.34], better on 20 tasks and worse on 4), and it was ahead of LoRA (31%) and full fine-tuning (27%) after 200 updates, although the difference to LoRA was not clear (+0.07 [−0.02, +0.17]). Plain k-NN, however, succeeded most often (44%); the difference to RAID v8 after 25 updates was −0.06 [−0.17, +0.05], better on 10 tasks and worse on 16 (Figure 6). These closed-loop comparisons were not pre-registered as hypotheses.

6.5 Offline error does not predict closed-loop success

k-NN had the highest offline error of the seven closed-loop conditions (0.526, MAE, seed 42) and the highest success; LoRA and full fine-tuning had the lowest offline error (0.293 and 0.287) and

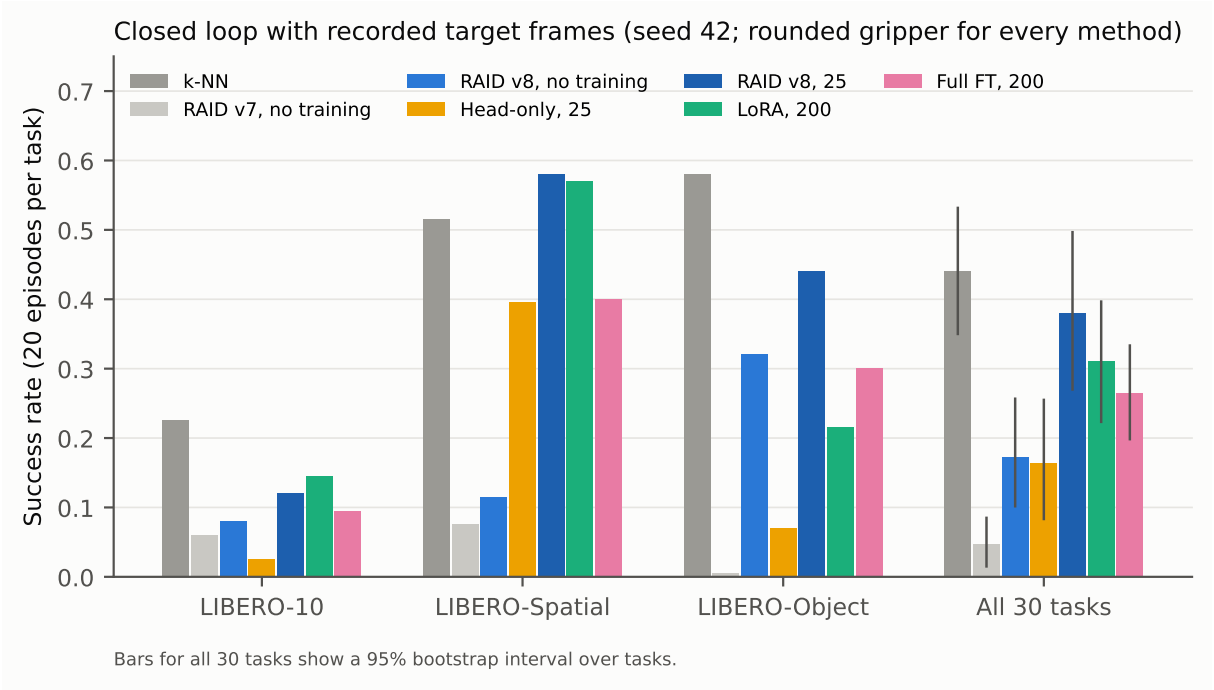


Figure 6: Closed-loop success per suite and over all 30 tasks for the seven conditions.

lower success than the adapted RAID v8 head (offline 0.337). Across the 210 task–method pairs, the rank correlation between lower offline error and higher success was −0.06, that is, none; within a task it averaged 0.16. Copying recorded actions keeps them physically consistent, and grasping depends on a few decisive steps that an average-error metric weights like any other.

7 Discussion and Limitations

What the fix achieves. A learned retrieval head can be worse than the nearest neighbour it is built on, because its gate cannot tell when its own prediction is wrong. Showing the gate the disagreement, and training it on cases where the prediction is wrong, removes this failure: without any training on the target task, the head was never worse than k-NN on any of 90 task–encoder combinations. The method adds 339,000 parameters to the head and 0.4 ms per update.

What it does not achieve. The fix improves the starting point of adaptation more than its end point. After updates, the head without the new inputs was slightly better, and fine-tuning the encoder reached the lowest error with 7 to 17 times the compute per update. Whether a schedule that turns corruption off during adaptation keeps both advantages is an open question for a new frozen study.

Offline error and success. Lower action error did not mean more finished tasks. Offline error remains useful to compare inverse-dynamics models under a fixed target, but claims about control need closed-loop evidence, and the closed loop favoured copying recorded actions.

Limitations.

- All tasks are simulated LIBERO tasks with five demonstrations each, and the closed-loop test uses recorded target frames, not a video planner.
- The design was developed on LIBERO-Goal tasks that our earlier study had already used; the final tasks were never used for design.
- LoRA, full fine-tuning and the closed-loop test use one seed and one encoder (MAE).

- The gripper rounding rule was chosen on development tasks after observing closed-loop failures; it is applied identically to all methods.
- We cannot verify that the pretrained encoders never saw LIBERO images, so we make no claim that the tasks are unseen by them.
- The bootstrap treats the 30 tasks as exchangeable; the three suites differ, and Table 4 reports them separately.

8 Conclusion

A learned gate between a parametric inverse-dynamics model and retrieved actions should know when the parametric prediction is wrong. Giving the gate the disagreement between the two predictions, and training it with corrupted predictions, made a retrieval-augmented inverse-dynamics model better than nearest-neighbour regression and than the same head without these inputs on all 30 sealed LIBERO tasks, for three different frozen encoders, without any training on the new task. With a few updates the head kept a small advantage over head-only fine-tuning for two of three encoders and finished more simulated tasks than head-only, LoRA and full fine-tuning, while encoder fine-tuning reached lower offline error and plain nearest-neighbour regression finished the most tasks.

Acknowledgments and Declarations

To be completed by the authors: funding, conflicts of interest, and data-use and licence statements for LIBERO and the pretrained encoders. Nothing is stated here until confirmed.

Reproducibility and Artifact Availability

All numbers come from collected, SHA-256-verified run outputs, computed by paper/v8_numbers.py:

- Final run (once): artifacts/raid_v8/final_20261023/ (amendment.json with digest `84375c3be0c0ddb953a34ccfa453a15f236aa7ae65ddb6139ed87382a05ee4b8`, collection/v8_final/ with all unit files and final_results.json).
- Development run: artifacts/raid_v8/dev_20261022/ (collection/v8_dev/, including the 27 source heads and normalizers bound by the amendment).
- Code: raid_experiments/v8/ (data splits, encoders, feature caches, heads, source training, target adaptation, closed loop, stage driver and analysis); design and hypotheses: docs/RAID_V8_PLAN.md.
- Figures: paper/figures/make_figures.py.

The data are the public LIBERO datasets; every downloaded file's SHA-256 is recorded with the run. Pretrained encoders are fixed by checkpoint hash (MAE) or Hugging Face revision (DINOv2, SigLIP).

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A Settings

Table A1. Settings shared by all methods unless stated.

Item	Value
Encoders	MAE ViT-B/16 (normalized CLS), DINOv2 ViT-B/14 (facebook/dinov2-base), SigLIP ViT-B/16 (google/siglip-base-patch16-224); all frozen, 768-d, 224 × 224 bicubic input
State	7 joint + 2 gripper positions; actions 8 × 7; both normalized with LIBERO-90 training statistics
Head trunk	MLP 512–256 with LayerNorm, ReLU, dropout 0.1
Retrieval	k = 16 (heads), k = 4 with temperature 0.1 (k-NN); distance as in Section 4.1 with exact duration match
RAID v8 gate	MLP with 256 hidden units on [state, confidence vector]; corruption probability 0.25
Selection loss	weight 0.1, temperature 1.0; candidate action jitter 0.05 during training
Source training	90 tasks; 4,000 updates; 8 tasks × 16 windows per batch; banks of 2/5/10 demonstrations; best validation checkpoint every 250 updates; seeds 42, 43, 44
Head adaptation	batch 64; AdamW, weight decay 10^{-4}, gradient clipping 1.0; checkpoints 0, 10, 25, 50, 100, 200
Encoder tuning	MAE only, seed 42; LoRA rank 8, α 16 on blocks 10–11; head learning rate $3{\cdot}10^{-4}$; checkpoints 0, 25, 50, 100, 200
REGENT-style blend	weight exp(−10 d) on the nearest support window, rest from the source head
Closed loop	20 query episodes per task; alignment window 24 frames; 10 settle steps; limit 1.5 × demonstration length + 20 steps; rounded gripper

B Development Results

Table B1. Development tasks (LIBERO-Goal, 10 tasks × 3 seeds), no target training: chunk MSE. The variant with the lowest MAE error was selected for the final run.

Method	MAE	DINOv2	SigLIP
VINN-style k-NN	0.592	0.471	0.506
REGENT-style blend	0.633	0.511	0.530
RAID v7	0.637	0.667	0.578
v8 gate without corruption	0.757	0.771	0.750
v8 gate with corruption (selected)	0.494	0.437	0.443

Table B2. Development tasks, after 25 updates at the selected learning rates: chunk MSE.

Method	MAE	DINOv2	SigLIP
Head-only fine-tuning	0.428	0.400	0.437
RAID v7 head	0.401	0.370	0.399
v8 gate without corruption	0.397	0.368	0.397
v8 gate with corruption (selected)	0.412	0.375	0.405

Development closed loop (raw gripper commands, before the rounding rule, MAE, 20 episodes $\times$ 10 tasks): k-NN 0.72, RAID v8 after 25 updates 0.57, LoRA after 200 updates 0.56, head-only after 25 updates 0.42, RAID v8 without training 0.34, RAID v7 without training 0.09, v8 gate without corruption 0.07.

C Run Record

The development stage ran four times. The first attempt found no free GPU; the second stopped while the pod booted; in the third, all stages except the closed loop finished, but the results were deleted with the cloud storage before they were collected (only the stage logs survived, and their numbers match the fourth run); the fourth run is the one reported. The final run ran once after one failed safety check of the cloud launcher; no final file was read before that run, and nothing was changed after it.